Brady (Bowei) Tian

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PROFILE

I am trying to build a theory of how machine intelligence represents and manipulates concepts — one that explains how raw inputs are organized into human-interpretable structure, and that uses this account to build agents whose problem-solving follows interpretable conceptual paths rather than opaque pattern-matching. The working thesis is that a network's representation space has measurable geometric structure: many concepts appear as approximately linear directions, and a suitably structured agent can compose these directions into a reasoning path toward a goal inferred from the user's input.

The program runs from analysis to system. Spectral Principal Paths (SPP) characterizes the conditions under which deep networks distill approximately linear concept representations from noisy inputs. On that basis I am building Mindcraft, a system that composes concept-aligned reasoning paths on demand. The main work stream aims to formalize how concepts form, propagate, and can be steered inside neural systems.

RESEARCH MAP

A theory of concept formation toward general intelligent agents

- **Concept geometry.** Formalizing how input-level information is amplified into high-level concepts (“honesty,” “fairness”) as stable, high-energy singular directions — unifying the Linear Representation Hypothesis and manifold views as special cases of a single framework.
- **From inspection to agency.** Building the bridge from *reading* a model’s internal concepts to *directing* them: an intelligence network that, given a user’s line of thought, selects a reasoning path and steers the model’s internal geometry to reach the intended answer quickly.
- **Tooling.** A standardized toolkit that traces stable spectral principal paths through hidden spaces from any input, giving an interpretable, geometric view of concept formation across DNNs, ViTs, and LLMs.

SELECTED WORK TOWARD THE THEORY

Spectral Principal Paths — a geometric reason of linear representation hypothesis

A Spectral Perspective on Linear Representation Formation

- Proposed the Input-Space Linearity Hypothesis: concept-aligned directions already reside in the raw input and are recovered and amplified during training, recasting training as a noise-suppression process along dominant spectral paths.
- Showed empirically that top singular values grow while their directions stay stable across training, so each input follows a consistent path toward its conceptual representation — validated across VLM and LLM settings.

Mindcraft — a theory merging linear representation and manifold

Composing concept-aligned reasoning paths conditioned on user intent

- Extending SPP from *description* to *decision*: a network that lays out a reasoning trajectory through concept space given a user’s existing line of thought, then drives the model along it to resolve the target problem efficiently.
- Connects concept inspection, steering, and counterfactual intervention into a single controllable pipeline.

INDUSTRY RESEARCH

Research Intern — Snowflake *Summer 2025*

- Designed a context-aware *TextGrad* auto-prompting framework that treats prompts as learnable parameters optimized with gradient-based updates informed by schema structure and execution feedback.

– Built a data-augmentation and lightweight LoRA fine-tuning pipeline letting prompts adapt to new databases; on the BIRD benchmark across 11 heterogeneous databases, achieved a 2% execution accuracy over the SOTA Text2SQL pipeline.

EARLIER RESEARCH — TRUSTWORTHY & SECURE AI

Before my research for interpretability theory, my work clustered around two trustworthy-AI themes that motivated the move toward a structural theory of concepts: fairness, causal inference, and the security of how models encode and can be manipulated.

Fairness of Representations — University of Maryland, College Park *2023 – 2025*

– Develop adaptive, back-propagation-hooked attention masking with a contrastive distance loss to improve the fairness–accuracy tradeoff in Vision Transformers — a 6.72% accuracy gain over leading methods at comparable fairness. *FairViT: Fair Vision Transformer via Adaptive Masking*, ECCV 2024.

– Subsequently, develop EXOgenous Causal reasoning (EXOC), a causal-inference framework to improve counterfactual fairness. *Towards Counterfactual Fairness through Auxiliary Variables*, ICLR 2025.

AI Security & Backdoors — Wuhan University *2022 – 2025*

– Design a QoE attack reaching success rates up to 82% higher than baselines at low poison ratios. *An Effective and Resilient Backdoor Attack Framework against Deep Neural Networks and Vision Transformers.* IEEE TDSC 2025.

– Develop a trigger scope strategy and manipulated the attention mechanism through Attention Diffusion to improve attack flexibility. *MEGATRON: Backdooring Vision Transformers with Invisible Triggers.* IEEE TDSC 2025.

EDUCATION

University of Maryland, College Park, MD, USA *Expected May 2028*

Ph.D. Student (3rd year), Electrical & Computer Engineering — advised by Dr. Ang Li

Wuhan University, Wuhan, China *June 2024*

B.Eng. in Information Security — Cumulative GPA: 3.9 / 4.0

SELECTED PUBLICATIONS

[1] B. Tian, Y. He, W. Ye, Z. Wang, M. Liu, A. Li. *MindCraft: How Concept Trees Take Shape In Deep Models. arXiv:2510.03265, 2025.*

[2] B. Tian, X. Lyu, M. Liu, H. Wang, A. Li. *Spectral Principal Paths: A Spectral Perspective on Linear Representation Formation in LLMs.* COLM 2026.

[3] B. Tian, Y. He, Z. Wang, M. Liu, Y. Wu, and A. Li. *SPARC: Scalable Path-Specific Counterfactual Fairness via Causal Conditional Independence.* ECCV 2026.

[4] B. Tian, Z. Wang, S. He, W. Ye, G. Sun, Y. Dai, Y. Wu, A. Li. *Towards Counterfactual Fairness through Auxiliary Variables.* ICLR 2025.

[5] B. Tian, R. Du, Y. Shen. *FairViT: Fair Vision Transformer via Adaptive Masking.* ECCV 2024.

[6] X. Gong*, B. Tian*, M. Xue, Y. Wu, Y. Chen, Q. Wang. *An Effective and Resilient Backdoor Attack Framework against Deep Neural Networks and Vision Transformers.* IEEE TDSC 2025.

[7] X. Gong, B. Tian, M. Xue, S. Li, Y. Chen, Q. Wang. *MEGATRON: Backdooring Vision Transformers with Invisible Triggers. IEEE TDSC 2025.*

SKILLS ETA.

Research interests: Interpretability & representation learning, concept geometry, causal reasoning, reasoning agents, LLMs & multimodal models, trustworthy AI.

Languages: C/C++, Python, SQL, HTML/CSS/JavaScript, MATLAB, Markdown.

Tools: PyTorch, TensorFlow, Git/GitHub, Docker, Conda, Jupyter, VS Code, LaTeX.

Honors: Lei Jun Scholarship (2024); Outstanding Student (2021, 2022, 2023).